

AM-Spaces With Unit Fail the SPAP in Infinite Dimension

Automated Math Discovery Smoke Test

June 8, 2026

Source and Open Question

This packet concerns Question 2 from:

Luis A. Garcia, Jose Lucas P. Luiz, Vinicius C. C. Miranda, *Norm attainment for multilinear operators and polynomials on Banach Spaces and Banach lattices*, arXiv:2605.12117.

The source location is Section 4, “Open Questions”, Question 2. The question appears on page 28 of the source paper:

Corollary 4.2. *Let $n \in \mathbb{N}$, E be a reflexive Banach lattice and F be a Dedekind complete Banach lattice. Consider the following implications:*

- (1) *Every $A \in \mathcal{L}^+(E^n; F)$ is norm-attaining.*
- (2) *Every $A \in \mathcal{L}^+(E^n; F)$ is weakly sequentially continuous.*
- (3) *Every $P \in \mathcal{P}^+(E^n; F)$ is weakly sequentially continuous.*
- (4) *Every $P \in \mathcal{P}^+(E^n; F)$ is norm-attaining.*

If E or F has the SPAP, then (1) $\iff$ (2) $\Rightarrow$ (3) $\iff$ (4).

In Section 3, we introduced the SPAP and presented examples of Banach lattices that do and do not have this property. So far, however, the only known class of Banach lattices with the SPAP consists of Banach lattices with order continuous norm whose order is given by a basis. This leads to the following question:

Question 2: Are there other examples of Banach lattices with the SPAP besides Banach lattices with order continuous norm whose order is given by a basis?

We showed in Section 3 that all positive bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$ are norm-attaining while there are non-norm attaining bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$. From this and our results, we obtain that all positive bilinear operators $\ell_4 \times \ell_4 \rightarrow L_1([0, 1])$ are weakly sequentially continuous while there are bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$ that are not weakly sequentially continuous. It would be interesting, if we get examples with the domain being non-atomic Banach lattices. In view of this, we leave the following question:

Question 3: Are all positive n -linear forms on $L_{p_1}[0, 1] \times \cdots \times L_{p_n}[0, 1]$ with $\sum_{i=1}^n \frac{1}{p_i} < 1$ weakly sequentially continuous?

The paper defines the sequentially positive approximation property (SPAP) as follows: a Banach lattice E has SPAP if there is a sequence $(S_n)_n \subset B_{\mathcal{F}_+(E;E)}$ such that

$$\lim_{n \rightarrow \infty} \| |S_n - \text{id}_E|(x) \| = 0$$

for every $x \in E^+$. The authors note that the known positive examples are Banach lattices with order-continuous norm whose order is given by a basis, and they ask whether there are other examples.

Claim

Theorem 1. *Let E be an infinite-dimensional AM-space with a strong unit $\mathbf{e}$, normalized so that $\|\mathbf{e}\| = 1$. Then E does not have the SPAP.*

Remark 1. *This is a partial negative result for Question 2. It does not construct a new Banach lattice with SPAP. Instead, it rules out the classical AM-space-with-unit family, including every infinite-dimensional $C(K)$ -space with its usual unit.*

Proof

We use the standard norm formula for positive operators on AM-spaces with unit.

Lemma 1. *Let E be an AM-space with unit $\mathbf{e}$. If $P : E \rightarrow E$ is positive, then*

$$\|P\| = \|P\mathbf{e}\|.$$

Consequently, if $R : E \rightarrow E$ is regular, then

$$\|R\| \leq \| |R|\mathbf{e} \|.$$

Proof. For $x \in B_E$ we have $|x| \leq \mathbf{e}$. If P is positive, then $|Px| \leq P|x| \leq P\mathbf{e}$, so $\|Px\| \leq \|P\mathbf{e}\|$. Taking the supremum over $x \in B_E$ gives $\|P\| \leq \|P\mathbf{e}\|$, while the reverse inequality is immediate from $\|\mathbf{e}\| = 1$.

For regular R , positivity of $|R|$ and the lattice inequality $|Rx| \leq |R|(|x|)$ give

$$\|Rx\| \leq \| |R|(|x|) \| \leq \| |R|\mathbf{e} \|$$

for every $x \in B_E$. Hence $\|R\| \leq \| |R|\mathbf{e} \|$. □

Assume toward a contradiction that E has the SPAP. Then there is a sequence of positive finite-rank operators $(S_n)_n \subset B_{\mathcal{F}_+(E;E)}$ such that

$$\| |S_n - \text{id}_E|(x) \| \longrightarrow 0 \quad (x \in E^+).$$

Applying this at the unit $\mathbf{e}$ gives

$$\| |S_n - \text{id}_E|(\mathbf{e}) \| \longrightarrow 0.$$

By the lemma,

$$\|S_n - \text{id}_E\| \leq \| |S_n - \text{id}_E|(\mathbf{e}) \| \longrightarrow 0.$$

Thus the identity operator on E is a norm limit of finite-rank operators.

Every norm limit of finite-rank operators is compact. Therefore id_E would be compact. But the identity operator on a Banach space is compact if and only if the closed unit ball is compact, equivalently if and only if the space is finite-dimensional. This contradicts the assumption that E is infinite-dimensional.

Hence no infinite-dimensional AM-space with unit has the SPAP.

Verification Notes

No numerical verification is needed for this packet. The proof reduces the SPAP condition at the order unit to operator-norm convergence of finite-rank operators to the identity, and then uses the elementary compactness obstruction for infinite-dimensional Banach spaces.